

Agricultural Crop Production Analysis

(1997–2021) using Tableau Dashboard

	 Entice How does someone initially become aware of this process?	 Enter What do people experience as they begin the process?	 Engage In the core moments in the process, what happens?	 Exit What do people typically experience as the process finishes?	 Extend What happens after the experience is over?
 Steps What does the person (or group) typically experience?	Identifies need for agricultural insights Looks for historical crop production data	Opens Tableau Dashboard Loads Agricultural Crop Production Data (1997–2021) Selects State, Crop, Year, and Season	Views KPI Cards (Production & Yield) Explores Production Trend (1997–2021) Analyzes State-wise Production & Yield Compares Crops and Cultivated Area Identifies Top Producing States & Crops	Interprets insights and draws conclusions Exports dashboard / downloads report	Uses insights for policies / research / farming / investment Monitors new data and updates trends periodically
 Interactions What interactions do they have at each step along the way? - Things: Digital touchpoints / physical objects - Places: Where are they? - People: Who do they see or talk to?	Agricultural Reports, News, Research Papers Government Portals, Agri Websites	Tableau Public Dashboard Filters (State, Crop, Year, Season)	Interactive India Map KPI Cards Line Charts (Trends) Bar Charts (State / Crop Comparison) Pie / Donut Charts Tooltips & Drill-Downs Data Table (View) Story Dashboard (Scenes) Legends, Colors & Annotations Calculated Fields (Yield = Production / Area) Dashboard Filters & Parameters Bookmarks & Highlights Present to Stakeholders	Export PDF / Image Share Dashboard Link Present to Stakeholders	Government Planning Meetings Research Publications Farming Decisions (Next Season) Investment Analysis
 Goals & motivations At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")	Help me understand agricultural performance Help me find reliable historical data	Help me access the right data quickly Help me focus on what matters	Help me compare states and crops Help me identify high-yield and high-production crops Help me find low-performing regions Help me understand seasonal & long-term trends Help me support data-driven decisions Help me reduce manual analysis	Help me make confident conclusions Help me present insights clearly	Help me improve future planning and strategies Help me stay updated with new trends
 Positive moments What steps does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?	Availability of long-term agricultural data Awareness of useful dashboards	Easy to open and navigate dashboard User-friendly filters	Clear KPI overview Interactive charts reveal patterns Easy comparison across states and crops Discovering top performing crops and regions Drill-down provides deeper insights Attractive visualisations	Quick report export Insights are easy to share	Helps in better planning and strategy Saves time in future analysis
 Negative moments What steps does a typical person find frustrating, confusing, angering, costly, or time-consuming?	Data scattered in multiple sources Not sure which data is reliable	Large dataset takes time to load Data in different units (Tonnes, Bales, Nuts)	Missing or null values in data Inconsistent data entries Too many dimensions to compare Difficult to find specific insights without filters Manual calculation is time-consuming Learning Tableau features takes time	Interpreting data incorrectly Export formatting issues	Data not updated regularly Need technical skills to update dashboards
 Areas of opportunity How might we make each step better? What ideas do we have? What have others suggested?	Provide a single trusted data source Awareness through webinars & training	Optimize data loading performance Add usage guidance / help section	Add predictive analytics & forecasting Integrate weather, soil & market price data Add data quality indicators & explanations Provide advanced filters & dynamic parameters Enable mobile-friendly dashboards Add AI-based crop recommendations	Automate report generation Provide templates for presentations	Schedule automatic data refresh Add alerts for new trends & anomalies